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(54) **Title**: POLYMER COMPOSITION COMPRISING A BLEND OF A MULTI-BLOCK THERMOPLASTIC ELASTOMER AND A POLYMER COMPRISING A GROUP 14 METAL

(57) **Abstract**: The present invention relates to a polymer composition comprising a blend of a multi-block thermoplastic elastomer comprising alternating hard polymeric blocks and soft polymeric blocks, and a polymer comprising a metal of Group 14 of the Periodic System of the Elements. The present invention also relates to a membrane comprising said blend and its application in separation processes, ion transport processes, ultrafiltration and nanofiltration.

Polymer composition comprising a blend of a multi-block thermoplastic elastomer and a polymer comprising a Group 14 metal

Field of the invention

The present invention relates to a polymer composition comprising a blend of a multi-block thermoplastic elastomer comprising alternating hard polymeric blocks and soft polymeric blocks, and a polymer comprising a metal of Group 14 of the Periodic System of the Elements (IUPAC Version 22 June 2007), i.e. silicon, germanium, tin or lead. The present invention further relates to membranes comprising said polymer composition and to the application of such membranes in separation processes, ion transport processes, ultrafiltration and nanofiltration.

Background of the invention

Multi-block polymers comprising hard polymeric blocks and soft polymeric blocks are well known in the art. The hard polymeric blocks provide thermoplastic properties to the multi-block polymer whereas the soft blocks provide elastomeric properties to the multi-block polymer. Such multi-block polymers are also known in the art as thermoplastic elastomers (cf. Kirk-Othmer, Encyclopedia of Chemical Technology, 3rd Ed., Vol. 7, pages 368 - 370, **1993**; Kirk-Othmer, Encyclopedia of Chemical Technology, 3rd Ed., Vol. 9, pages 15 - 37, **1994**). Thermoplastic elastomers are used in a multitude of applications including membranes which are used in separation processes.

Thermoplastic elastomers are *inter alia* used in separation processes where they are applied in the form of membranes. Examples of commercially available thermoplastic elastomers used for this purpose include Pebax® grades, Pellethane® grades, Arnitel® grades and Hytrel® grades. All those grades have polyether blocks, amorphous polyester blocks or polysiloxane blocks as the soft blocks which implies that their major difference lies in the hard blocks. Pebax® grades, for example, are based on polyamide hard blocks. Pellethane® grades are based on polyurethane hard blocks. Arnitel® grades and Hytrel® grades are based on non-amorphous polyester blocks. Other hard blocks are based on imides.

US 4.963.165, incorporated by reference, discloses a membrane for gas separation which comprises a multi-block polyamide-polyether polymer.

US 5.130.017, incorporated by reference, discloses a membrane for separating aromatic compounds from non-aromatic compounds. The membrane is manufactured from a multi-block polymer comprising an amide acid prepolymer which is chain-extended with a second prepolymer selected from the group of prepolymers comprising dianhydride, its corresponding tetra-acid or diacid-diester and an epoxy component, a diisocyanate component or a polyester component.

US 5.290.452, incorporated by reference, discloses polyester-polyamide based membranes for separating aromatic compounds from non-aromatic compounds.

DE 4237604 A1, incorporated by reference, discloses a membrane comprising a polyamide-polyether block copolymer which is used for ultrafiltration.

Silicon containing polymers have been used for manufacturing membranes which find application in separation processes. For example, JP A 61118412 and US 5.494.989, both incorporated by reference, disclose a copolymer suitable for gas separation, said copolymer being synthesized by copolymerizing e.g. a silylarylacetylene and another acetylene monomer in the presence of a Group 5 metal catalyst.

US 5.707.423, incorporated by reference, discloses a separation membrane based on poly(1-dimethylphenylsilyl-1-propyne) for separating C^{2+} from natural gas.

US 4.567.245, incorporated by reference, discloses copolymer synthesised by copolymerising 1-trimethylsilyl-1-propyne and 1-(1',1',3',3'-tetramethyl-1',3'-disilabutyl)-1-propyne and a membrane made thereof having an outstandingly large oxygen permeation coefficient and a large separation factor between oxygen and nitrogen.

JP A 1009208, US 4.657.564 and US 5.373.073, all incorporated by reference, discloses membranes based on 1-alkenyldimethylsilyl-1-propyne.

JP A 1194903, incorporated by reference, discloses a liquid separating membrane which can efficiently separate water-alcohol mixtures, wherein the separating membrane is based on a poly(1-trimethylsilyl-2-trialkylsilyl-1-propyne).

US 5.449.728, incorporated by reference, discloses a membrane based on 1-[dimethyl(10-pinanyl)silyl]-1-propyne which is used in resolving optically active compounds.

RU 2206773 C1, incorporated by reference, discloses the synthesis of poly(5-trimethylsilylnorborn-2-ene) and its use in membranes for gas separation.

T.C. Merkel et al., J. Membr. Sci. *191*, 85 – 94, **2001**, incorporated by reference, discloses membranes made of poly(dimethylsiloxane) and poly(1-trimethylsilyl-1-propyne).

M. Ghisellini *et al.*, Desalination *149*, 441 – 445, **2002**, incorporated by reference, discloses membranes made of pure poly(1-trimethylsilyl-1-propyne) and of copolymers of 1-trimethylsilyl-1-propyne and 1-trimethylsilyl-1-hexyne and the mass transport properties of n-pentane and n-hexane in these membranes.

A. Car *et al.*, J. Membr. Sci. *307*, 88 – 95, and 110 – 117, **2008**, incorporated by reference, discloses membranes made of a blend of the polyamide-polyether block polymer Pebax® MH 1657 and polyethylene glycol having a M_n of 200 and their application in gas separation, in particular of carbon dioxide. Although the separation performance of these membranes was better than of normal polyamide-polyether block polymer membranes, their is still a need in the art for membranes having an improved separation performance.

US 2002/0028156, incorporated by reference, discloses a blood oxygenator membrane comprising a polymeric matrix and a reinforcing fibre-based material, wherein the polymeric matrix comprises a thermoplastic elastomer. The thermoplastic elastomer may be selected from a large group of homopolymers and copolymers. According to the examples, preferred thermoplastic elastomers are the polymers Tecoflex®, Hytrel® and Kraton®.

US 5.552.483, incorporated by reference, discloses a curable composition comprising a natural or synthetic rubber and a block polymer comprising alternating blocks of a polysiloxane and a copolymer of a 1,3-conjugated diene and a monovinyl aromatic compound. The polysiloxane block is preferably an elastomeric linear polysiloxane.

Although the membranes known from the prior art may have an acceptable performance, the present inventors have found that the addition of “smart” additives to the base polymeric constituent used for the manufacture of the membrane enhances its properties, in particular separation properties, ion transport processes, ulttrafiltration and nanofiltration. In addition, it appears that membranes known from the prior art have widely different properties and that they can be used in very different separation processes, despite the fact that they are based on similar polymeric substances.

Summary of the invention

The present invention relates to a polymer composition comprising a blend of a multi-block thermoplastic elastomer comprising alternating hard polymeric blocks and soft polymeric blocks, and a polymer comprising a metal of Group 14 of the Periodic System of the Elements (IUPAC Version 22 June 2007).

The present invention further relates to a membrane for separation processes, wherein said membrane comprises a polymer composition comprising a blend of a multi-block thermoplastic elastomer comprising alternating hard polymeric blocks and soft polymeric blocks, and a polymer comprising a metal of Group 14 of the Periodic System of the Elements.

The present invention also relates to a gas separation process involving a membrane comprising a polymer composition comprising a blend of a multi-block thermoplastic elastomer comprising alternating hard polymeric blocks and soft polymeric blocks, and a polymer comprising a metal of Group 14 of the Periodic System of the Elements.

The present invention relates in particular to a carbon dioxide gas separation process involving a membrane comprising a polymer composition comprising a blend of a multi-block thermoplastic elastomer comprising alternating hard polymeric blocks and soft polymeric blocks, and a polymer comprising a metal of Group 14 of the Periodic System of the Elements.

The present invention further relates to ion transport processes which occur in e.g. ion exchange, electrochemical cells including batteries, and fuel cells.

The present invention also relates to ultrafiltration and nanofiltration processes.

Detailed description of the invention

The verb “to comprise” as is used in this description and in the claims and its conjugations is used in its non-limiting sense to mean that items following the word are included, but items not specifically mentioned are not excluded. In addition, reference to an element by the indefinite article "a" or "an" does not exclude the possibility that more than one of the element is present, unless the context clearly requires that there is one and only one of the elements. The indefinite article "a" or "an" thus usually means "at least one".

The multi-block thermoplastic elastomer comprising alternating hard polymeric blocks and soft polymeric blocks have the general formula $-(A-B)_n-$ wherein A represents a hard polymeric block and wherein B represents a soft polymer block and wherein n represents an integer of at least 2.

The multi-block thermoplastic elastomer

According to the present invention, the multi-block thermoplastic elastomer comprises alternating hard polymeric blocks and soft polymeric blocks. Preferably, the polymeric hard blocks are selected from the group consisting of polyurethane blocks, polyamide blocks, non-amorphous polyester blocks, polyimide blocks, polysulfone blocks, polycarbonate blocks and mixtures thereof. Also preferably, the soft polymeric blocks are selected from the group consisting of polyether blocks, amorphous polyester blocks and polysiloxane blocks. The present invention therefore encompasses multi-block thermoplastic elastomers according to the general formula $-(A-B)_n-$ wherein A represents a hard polymeric block and wherein B represents a soft polymer block in the following combinations:

A = a polyurethane block and B = a polyether block, an amorphous polyester block, a polysiloxane block, or mixtures thereof;

A = a polyamide block and B = a polyether block, an amorphous polyester block, a polysiloxane block, or mixtures thereof;

A = non-amorphous polyester block and B = a polyether block, an amorphous polyester block, a polysiloxane block, or mixtures thereof;

A = a polyimide block and B = a polyether block, an amorphous polyester block, a polysiloxane block, or mixtures thereof;

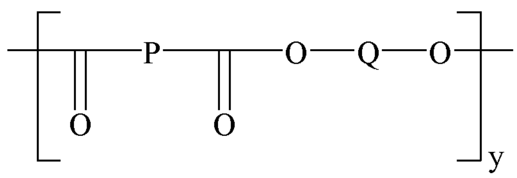
A = a polysulfone block and B = a polyether block, an amorphous polyester block, a polysiloxane block, or mixtures thereof; and

A = a polycarbonate block and B = a polyether block, an amorphous polyester block, a polysiloxane block, or mixtures thereof.

According to the present invention, it is preferred that the soft blocks have a Tg of 0°C or lower.

More preferably, the multi-block thermoplastic elastomer comprises a polyamide block as the hard block and a polyether block, an amorphous polyester block or a polysiloxane block as the soft block. Most preferably, the multi-block thermoplastic elastomer comprises a polyamide block as the hard block and a polyether block as the soft block.

According to a first embodiment of the present invention, the multi-block thermoplastic elastomer comprising a polyamide block as the hard block and a polyether block as the soft block has preferably the following general Formula (VII):



Formula (VII)

wherein P is a polyamide block, O is oxygen and Q is a polyether block whereas y indicates an integer such that the intrinsic viscosity η of the multi-block thermoplastic elastomer is in the range of about 0.5 to about 2.5, preferably about 0.7 to about 2.2, more preferably about 0.8 to about 2.05.

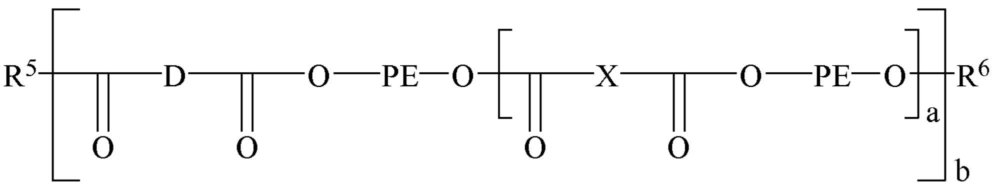
The polyamide block has preferably a number average molecular weight of about 100 to about 25000, more preferably about 100 to about 15000 and most preferably about 500 to about 10000. The polyamide block is preferably an aliphatic polyamide block. Most preferably, the polyamide block is polyamide 6 (PA6).

The polyether block has preferably a number average molecular weight of about 100 to about 10000, more preferably about 200 to about 6000 and most preferably about 400 to about 3000. The polyether block is preferably a polyoxyalkylene glycol block such as a polyoxyethylene glycol block, a polyoxypropylene glycol block (which may be derived from 1,2-propene oxide, 1,3-propene oxide or mixtures thereof), a poly(oxyethylene-*co*-oxypropylene) glycol block, a polyoxytetramethylene glycol block (also known as poly-THF) or mixtures thereof. Most preferably, the polyethylene glycol block is a polyoxyethylene glycol block.

The weight proportions of the polyether block to the total weight of the multi-block thermoplastic elastomer having polyamide blocks as hard blocks is preferably about 5 to about 85 wt.%, more preferably about 20 wt.% to about 75 wt.%, most preferably about 40 wt.% to about 70 wt.%.

The multi-block thermoplastic elastomers according to the first embodiment of the present invention and methods for their preparation are for example disclosed in US 4.230.838, US 4.331.786, US 4.332.920 and US 4.376.856, all incorporated by reference.

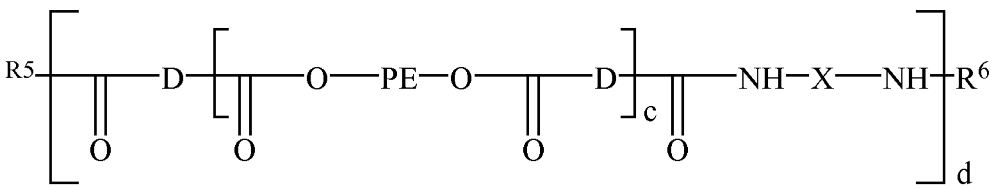
According to a second embodiment of the present invention, the multi-block thermoplastic elastomer comprising a polyamide block as the hard block and a polyether block as the soft block has preferably the following general Formula (VIII):



Formula (VIII)

wherein R^5 and R^6 are OH and/or H, a is a number 0.1 to 10, b is a number of 2 to 50, D is a residue of an oligoamidediacid having a M_n of about 300 to about 8000, PE is a residue of a polyoxyalkylene, preferably a polyoxyethylene, having a M_n of about 200 to about 5000, and X is a residue of a diacid comprising linear or branched aliphatic, cycloaliphatic or aromatic hydrocarbon residues having 4 to 12 carbon atoms. Such multi-block thermoplastic elastomers are disclosed in US 5.166.309, incorporated by reference. D may comprise a residue of a diacidic limiter such as dodecanedioic acid.

According to a third embodiment of the present invention, the multi-block thermoplastic elastomer comprising a polyamide block as the hard block and a polyether block as the soft block has preferably the following general Formula (IX):



Formula (IX)

wherein R^5 and R^6 are OH and/or H, c is a number 1 to 4, d is a number of 2 to 50, D is a residue of an oligoamidediacid having a M_n of about 300 to about 3000, PE is a residue of a polyoxyalkylene, preferably a polyoxyethylene, having a M_n of about 200 to about 5000, and X is a residue of a diacid comprising linear or branched aliphatic, cycloaliphatic or aromatic hydrocarbon residues having 4 to 20 carbon atoms. Such multi-block thermoplastic elastomers are disclosed in US 5.213.891, incorporated by reference.

Additionally, the multi-block thermoplastic elastomer according to the three embodiments mentioned above comprising a polyamide block as the hard block and a polyether block as the soft block may further comprise a polyol as is disclosed in US 2007/0106034, incorporated by reference, or a substituted polyoxyalkylene glycol block as is disclosed in US 6.300.463, incorporated by reference.

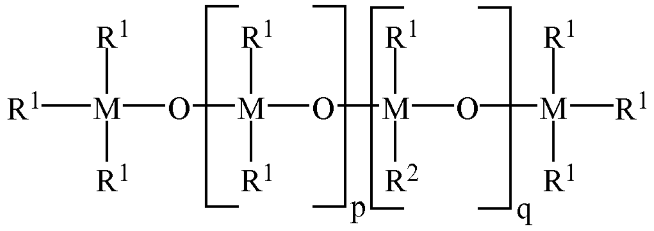
According to the present invention, the multi-block thermoplastic elastomer comprising a polyamide block as the hard block and a polyether block as the soft block are most preferably selected from the PEBAX® grades 2533 SA 01, 2533 SD 01, 2533 SN 01, 3533 SA 01, 3533 SD 01, 3533 SN 01, 4033 SA 01, 4033 SD 01, 4033 SN 01, 5533 SA 01, 5533 SN 01, 5533 SP 01, 6633 SA 01, 6633 SP 01, 7033 SA 01, 7033 SP 01, 7233 SA 01, 7233 SP 01, MH 1657, MP 1878 SA 01, MV 1041 SA 01, MV 1074 SP 01, MV 3000 SA 01, MV 6100 SA 01, MX 1205 SA 01, MX 1205 SP 01, MX 1717, RDG 277 and RDG 314.

<u>The polymer comprising a metal of Group 14 of the Periodic System of the Elements</u>

Group 14 of the Periodic System of the Elements contains the metals silicon, germanium, tin and lead. According to the present invention, it is preferred that the Group 14 metal is silicon (Si).

The polymer comprising the metal of Group 14 of the Periodic System of the Elements can be selected from the group consisting of:

(a) a polysiloxane according to Formula (I):



Formula (I)

wherein:

M is the Group 14 metal;

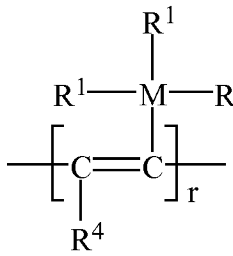
R^1 is selected from the group consisting of linear or branched C_1 - C_{12} alkyl groups, C_6 - C_{12} aryl groups and mixtures thereof;

R^2 is a polyoxyalkylene chain;

p is in the range of 5 to 5000; and

q is in the range of 5 to 10000;

(b) a polyalkyne according to Formula (II):



Formula (II)

wherein:

M is the Group 14 metal;

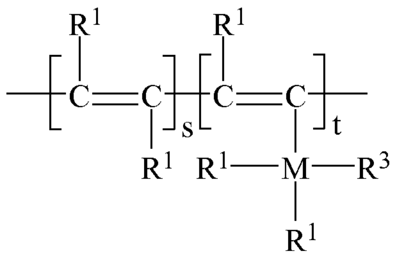
R^1 is selected from the group consisting of linear or branched C_1 - C_{12} alkyl groups, C_6 - C_{12} aryl groups and mixtures thereof;

R^3 is R^1 or a polyoxyalkylene chain;

R^4 is hydrogen, a linear C_1 - C_{12} alkyl group, a C_6 - C_{12} aryl group, a $(R^1)_3M$-CH_2- group, a $(R^1)_2R^3M$-CH_2- group, or a mixture thereof; and

r is in the range of 100 – 20000;

(c) a polyalkyne according to Formula (III):



Formula (III)

wherein:

M is the Group 14 metal;

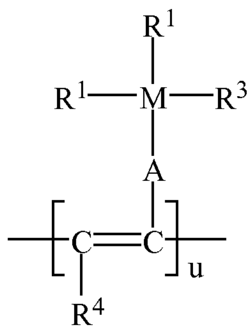
R^1 is selected from the group consisting of linear or branched C_1 - C_{12} alkyl groups, C_6 - C_{12} aryl groups and mixtures thereof;

R^3 is R^1 or a polyoxyalkylene chain;

$(s/(s+t)) \geq 0.2$;

$(t/(s+t)) \geq 0.05$;

(d) a polyalkyne according to Formula (IV):



Formula (IV)

wherein:

M is the Group 14 metal;

A is $-(CR^1{}_2)_v-$ wherein $v = 1 – 6$;

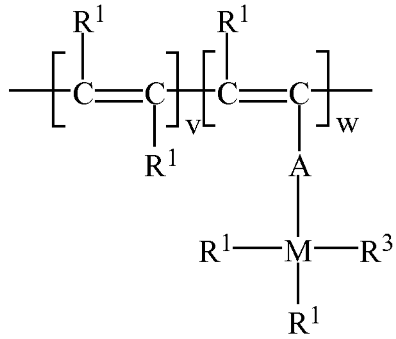
R^1 is selected from the group consisting of linear or branched C_1 - C_{12} alkyl groups, C_6 - C_{12} aryl groups and mixtures thereof;

R^3 is R^1 or a polyoxyalkylene chain;

R^4 is hydrogen, a linear C_1 - C_{12} alkyl group, a C_6 - C_{12} aryl group, $(R^1)_3M-CH_2-$ group, a $(R^1)_2R^3M-$ group, or a mixture thereof; and

u is in the range of 100 – 20000;

(e) a polyalkyne according to Formula (V):



Formula (V)

wherein

M is the Group 14 metal;

A is $-(CR^1_2)_v-$ wherein v = 1 – 6;

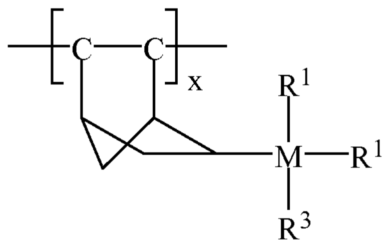
R^1 is selected from the group consisting of linear or branched C_1 - C_{12} alkyl groups, C_6 - C_{12} aryl groups and mixtures thereof;

R^3 is R^1 or a polyoxyalkylene chain;

$(v/(v+w)) \geq 0.2$; and

$(w/(v+w)) \geq 0.05$; or

(f) a polyalkene according to Formula (VI);



Formula (VI)

wherein R^1 is selected from the group consisting of linear or branched C_1 - C_{12} alkyl groups, C_6 - C_{12} aryl groups and mixtures thereof;

R^3 is R^1 or a polyoxyalkylene chain; and

x is in the range of 100 – 20000.

In the polymers (a) – (f) disclosed above, any one of the groups R^1, R^2, R^3 and R^4 may contain a chiral centre. In addition, the groups bonded to the metal M may be all different thereby rendering chirality to the metal M.

In the polymers (a) – (f), any one of the groups R^1, R^2, R^3 and R^4 may be substituted with a substituent selected from the group of linear and branched C_1 –C_6 alkyl, halogen, OH, $-OR^1$, $M(R^1)_3$ or $M(R^1)_2R^3$.

In the polymers (a) – (f) disclosed above, it is preferred that R^1 a linear C_1 – C_6 alkyl group or phenyl, more preferably a C_1 – C_6 alkyl group, most preferably methyl.

It is also preferred that R^3 has the same meaning as R^1.

In the polymers (a) – (f) disclosed above, it is furthermore preferred that the polyoxyalkylene chain (R^2 and/or R^3) is a polyoxyethylene chain, said polyoxyethylene chain preferably having the formula:

$$-(CH_2CH_2CH_2O)-[-CH_2CH_2O-]_p-CH_3$$

wherein p is in such a range that the M_n of the polyoxyethylene chain is equal to about 1000 or less, more preferably equal to about 500 or less, and most preferably equal to about 250 or less. The minimum M_n is 117 (i.e. p = 1).

In the polymers (a) – (f) disclosed above, it is preferred that R^4 is hydrogen, methyl or phenyl, most preferably methyl or phenyl.

In the polymers (a) – (f) disclosed above, it is furthermore preferred that the weight average molecular weight M_w of polymers (a) – (f), i.e. the polysiloxane according to Formula (I) (wherein R^2 is the polyoxyalkylene chain), the polyalkynes according to Formula (II), (III), (IV) or (V), or the polyalkene according to Formula (VI) (i.e., that if R^3 is a polyoxyalkylene chain) is determined by the polyoxyalkylene chain for about 60% to about 90%, more preferably about 70 wt.% to about 85 wt.%.

Furthermore, it is known in the prior art that poly(trialkylsilylalkynes) have generally a high molecular weight, i.e. a M_w over 10^6. Reference is made to US 4.808.679, incorporated by reference.

More specific preferred embodiments of polymers (b) – (f) are given below.

In polymer (b), R^1 and R^3 are most preferably methyl and R^4 is most preferably a linear C_1 – C_4 alkyl group. Most preferably, R^4 is methyl so that polymer (b) is poly(1-trimethylsilyl-1-propyne) (PTMSP).

In polymer (c), R^1 and R^3 are most preferably methyl. It is furthermore preferred that the monomer bearing the $-M(R^1)_2R^3$ substituent is present in the final polymer (c) in an amount of about 5 wt.% to about 80 wt.%, based on the total weight of the polymer (c), and that the other monomer is present in an mount of about 20 wt.% to about 95 wt.%, based on the total weight of the final polymer (c).

In polymer (d), R^1 and R^3 are most preferably methyl and R^4 is most preferably a linear $C_1 – C_4$ alkyl group. Most preferably, R^4 is methyl.

In polymer (e), v is preferably 1 and R^1 and R^3 are most preferably methyl.

In polymer (f), R^1 and R^3 are most preferably methyl.

The blend of the multi-block thermoplastic elastomer and the polymer comprising a metal of Group 14 of the Periodic System of the Elements

According to the present invention, it is preferred that the blend comprises about 0.1 wt.% to about 99 wt.% of the polymer comprising a metal of Group 14 of the Periodic System of the Elements, based on the total weight of the blend. It is furthermore preferred that the blend comprises about 1 wt.% to about 99.9 wt.% of the multi-block thermoplastic elastomer comprising alternating hard polymeric blocks and soft polymeric blocks, based on the total weight of the blend. More preferably, the blend comprises about 0.1 wt.% to about 97 wt.% of the polymer comprising a metal of Group 14 of the Periodic System of the Elements and about 3 wt.% to about 99.9 wt.% of the multi-block thermoplastic elastomer comprising alternating hard polymeric blocks and soft polymeric blocks. Even more preferably, the blend comprises about 1 wt.% to about 97 wt.% of the polymer comprising a metal of Group 14 of the Periodic System of the Elements and about 3 wt.% to about 99 wt.% of the multi-block thermoplastic elastomer comprising alternating hard polymeric blocks and soft polymeric blocks. Yet even more preferably, the blend comprises about 10 wt.% to about 95 wt.% of the polymer comprising a metal of Group 14 of the Periodic System of the Elements and about 5 wt.% to about 90 wt.% of the multi-block thermoplastic elastomer comprising alternating hard polymeric blocks and soft polymeric blocks.

The polymer composition

The polymer composition preferably comprises about 50 wt.% to about 100 wt.% of the blend of a multi-block thermoplastic elastomer comprising alternating hard polymeric blocks and soft polymeric blocks, and a polymer comprising a metal of

Group 14 of the Periodic System of the Elements, based on the total weight of the polymer composition. More preferably, the polymer composition comprises about 70 wt.% to about 100 wt.% of said blend. Yet even more preferably, the polymer composition comprises about 90 wt.% to about 100 wt.% of said blend. Most preferably, the polymer composition consists essentially of said blend.

The polymer blend may comprise, as a further component, another thermoplastic elastomer, preferably a multi-block thermoplastic elastomer, in particular a multi-block thermoplastic elastomer selected from the group consisting of multi-block thermoplastic elastomers according to the general formula $-(A-B)_n-$ wherein A represents a hard polymeric block and wherein B represents a soft polymer block in the following combinations:

A = a polyurethane block and B = a polyether block, an amorphous polyester block, a polysiloxane block, or mixtures thereof;

A = a polyamide block and B = a polyether block, an amorphous polyester block, a polysiloxane block, or mixtures thereof;

A = non-amorphous polyester block and B = a polyether block, an amorphous polyester block, a polysiloxane block, or mixtures thereof;

A = a polyimide block and B = a polyether block, an amorphous polyester block, a polysiloxane block, or mixtures thereof;

A = a polysulfone block and B = a polyether block, an amorphous polyester block, a polysiloxane block, or mixtures thereof; and

A = a polycarbonate block and B = a polyether block, an amorphous polyester block, a polysiloxane block, or mixtures thereof.

The amount of this other thermoplastic elastomer is preferably about 0 wt.% to about 50 wt.%, based on the total weight of the polymer composition. More preferably, this amount is about 0 wt.% to about 30 wt.%, even more preferably about 0 wt.% to about 10 wt.% and most preferably about 0 wt.%.

The polymer composition according to the present invention is in particular suitable for the manufacture of a membrane. When any of the polymers (a) – (f) comprise an unsaturated carbon carbon bond, these polymers are preferably crosslinked to provide additional mechanical strength to the membrane.

The membrane

The membrane according to the present invention is preferably used in various applications which include separation processes, ion transport processes, in particular processes involving the transport of Li^+, ultrafiltration processes and nanofiltration processes. Additionally, since the polymers (a) – (f) may comprise a chiral centre, the membrane according to the present invention may be suitable for the separation of optical isomers, in particular enantiomers, as is for example disclosed in US 5.449.728, incorporated by reference.

The separation processes include gas/gas separation processes, gas/liquid separation processes and liquid/liquid separation processes. More preferably, the separation process is a gas/gas separation process, wherein most preferably carbon dioxide is separated from another gas or a mixture of gases. Even more preferably, the gas/gas separation process involves the separation of carbon dioxide from a light gas, wherein the molecular weight of the light gas is 44 or less (the molecular weight of propane is 44). Most preferably, the gas/gas separation process involves the separation of carbon dioxide from a gas which comprises hydrogen, nitrogen, methane or a mixture thereof, and wherein the molecular weight of the heaviest gas present in the mixture is 44 or less.

The membrane according to the present invention has preferably a CO_2 gas permeability coefficient (determined at 35°C and 400 kPa pressure) of at least 120 Barrer, more preferably at least 140 Barrer, yet even more preferably at least 150 Barrer and most preferably at least 170 Barrer (Barrer = 10^{-10} cm^3 (STP).cm/(cm^2.s.cm Hg = $7.5 \cdot 10^{-18}$ $m^3 \cdot m/m^2 \cdot s \cdot Pa$).

The membrane according to the present invention has preferably a hydrogen gas permeability coefficient (determined at 35°C and 400 kPa pressure) of less than 17 Barrer.

The membrane according to the present invention has preferably a nitrogen gas permeability coefficient (determined at 35°C and 400 kPa pressure) of less than 5 Barrer.

The membrane according to the present invention has preferably a methane gas permeability coefficient (determined at 35°C and 400 kPa pressure) of less than 12 Barrer.

The permeability coefficient P is herein defined as:

$$P = D \times S$$

wherein D is the diffusivity ($cm^2/s = 1{\cdot}10^{-4}\ m^2/s$) and S is the solubility (cm^3 (STP)/(cm^3.cm Hg = $7.5{\cdot}10^{-4}\ m^3/m^3{\cdot}Pa$).

The membrane according to the present invention has preferably a selectivity $\alpha[P(CO_2)/P(H_2)]$ of at least about 9.0. The selectivity $\alpha[P(CO_2)/P(N_2)]$ is preferably at least about 30. As a consequence, the membrane according to the present invention provides a strongly enhanced permeability for CO_2 gas in combination with an enhanced selectivity for CO_2 gas over nitrogen gas and/or hydrogen gas.

According to the present invention, the selectivity $\alpha[P(CO_2)/P(CH_4)]$ of the membrane is preferably at least about 10. Additionally, the selectivity $\alpha[P(CO_2)/P(O_2)]$ is preferably at least about 15 and the selectivity $\alpha[P(CO_2)/P(He)]$ is preferably at least about 15.

The selectivity $\alpha[P(A)/P(B)]$ is herein defined as:

$$\alpha[P(A)/P(B)] = \frac{P(A)}{P(B)}$$

wherein P(A) and P(B) represent the permeability coefficients of gases A and B, respectively.

Preferably, the membrane according to the present invention further comprises a support for providing mechanical strength. Such supports are known in the art and are generally highly permeable and occur in the form of flat sheets, hollow fibers or hollow tubes. The support may be constituted of porous glass, porous metal, porous ceramics, polymers having a high permeability and the like. The membrane according to the present invention may be applied on one surface or on both surfaces of the support.

The present invention also relates to the use of the membrane according to the present invention in separation processes and ion transport processes, in particular processes involving the transport of Li^+, ultrafiltration processes and nanofiltration processes.

Examples

Example 1

Pebax® MH 1657 was obtained from Arkema. This thermoplastic elastomer is a block polymer containing about 60 wt.% PEO and about 40 wt.% PA-6. The M_n of the PEO block is about 1500 which means that the copolymer consists of about 35 repeating ethylene oxide monomers and about 9 repeating PA-6 monomers. The additive poly(dimethylsiloxane-*graft*-ethyleneglycol) (PDMS-PEG) was obtained from Aldrich and contains about 80 wt.% of PEG; the M_n is about 3800.

Membranes were prepared by the following general procedure. A solution of 3 wt.% Pebax® MH 1657 in a mixture of ethanol/water (70/30 w/w) was prepared under reflux at 80°C under continuous stirring (about two hours). After complete dissolution of the polymer, the solution was cooled to ambient temperature. Subsequently, different amounts of PDMS-PEG were added and the solution was stirred for one more hour. Membrane films were prepared by solution casting: the solution was poured in a petri dish and placed under a nitrogen atmosphere at ambient temperature to evaporate the solvent. The obtained films were dried in a vacuum oven at 30°C to remove residual solvent.

Thermal properties of the films were determined by DSC (Perkin Elmer DSC 7). The glass transition temperature T_g is defined as the midpoint of the heat capacity transition of the heating scan and the melting temperature from the onset of melting.

The melting enthalpies ΔH_m of the hard and soft blocks were determined from the endothermic peak areas. Samples were heated from -80°C to 250°C at a rate of 20°C/min. Subsequently, a cooling scan from 250°C to -80°C at a rate of 20°C/min followed by a second heating scan under the same conditions as the firs heating scan was performed. Results are shown in Table 1.

Table 1

Sample No.	PDMS-PEG Fraction (wt.%)	T_g (°C)	T_m (PEO) (°C)	ΔH_m (PEO) (J/g PEO)	T_m (PA-6) (°C)	ΔH_m (PA-6) (J/g PA-6)
1	0	-51.0	15.3	41	205	73
2	11	-50.9	18.3	45	204	75
3	21	-52.0	17.3	50	206	71
4	30	-51.4	17.6	59	203	74
5	40	-52.4	15.9	67	203	74
6	50	-51.1	15.8	78	203	68

From the date shown in Table 1, it can be concluded that PDMS-PEG has only a minor influence on ΔH_m (PA-6). The T_g of the blends is about constant ($\approx$ -51°C).

Example 2

Gas permeabilities of flat membrane films were determined with different gas-permeation set-ups. Low pressure gas (N_2, O_2, He, H_2, CH_4 and CO_2) permeabilities were determined with a low pressure gas separation (LPGS) set-up at 400 kPa feed pressure at 35°C. High pressure pure and mixed gas permeabilities were determined with a high pressure gas separation (HPGS) set-up for pressure up to 2500 kPa at 35°C. Bot set-ups measured the steady-state pressure increase in time in a calibrated volme at the permeation side of the membrane film (start: vacuum < 10 kPa) following the constant volume variable pressure method (A. Bos *et al.*, Separation and Purification Technology, *14(1-3)*, 27 – 39, **1998**). Results are shown in Table 2.

Table 2

Sample No.	P_{CO_2} (Barrer)	P_{H_2} (Barrer)	P_{N_2} (Barrer)	P_{CH_4} (Barrer)	P_{O_2} (Barrer)	P_{He} (Barrer)
1	98	10.2	1.8	6.0	4.7	6.1
2	179	18.1	3.7	12.4	9.3	10.5
3	245	23.9	5.8	18.9	13.6	13.5
4	326	31.1	8.1	27.3	19.1	17.1
5	457	43.4	12.2	40.7	27.8	23.5
6	532	50.2	14.8	49.4	33.5	26.7

The data shown in Table 2 show that the permeability increases significantly with increasing amount of PDMS-PEG.

Example 3

The pure gas permeability selectivities of CO_2 over several light gases are presented in Table 3 (determined with LPGS set-up).

Table 3

Sample No.	$\alpha(P_{CO_2}/ P_{H_2})$	$\alpha(P_{CO_2}/ P_{N_2})$	$\alpha(P_{CO_2}/ P_{CH_4})$	$\alpha(P_{CO_2}/ P_{O_2})$	$\alpha(P_{CO_2}/ P_{He})$
1	9.5	53.2	16.1	20.8	15.9
2	9.9	48.0	14.5	19.2	17.1
3	10.3	42.6	13.0	18.0	18.1
4	10.5	40.1	12.0	17.1	19.1
5	10.5	37.4	11.2	16.4	19.4
6	10.6	36.1	10.8	15.9	19.9

Since an increase in permeability is more desirable, even a slightly decreasing selectivity is acceptable. The data of Tables 2 and 3 demonstrate that the permeability of CO_2 can be increased significantly without significantly deteriorating the selectivity of CO_2 over light gases.

Comparative Examples 1 - 5

The CO_2 permeability of a membrane made from a blend of 50 wt.% Pebax® MH 1657 and 50 wt.% PEG 200 is 172 Barrer. The $\alpha(P_{CO_2}/P_{H_2})$ is 10.5, the $\alpha(P_{CO_2}/P_{N_2})$ is 50.5 and the $\alpha(P_{CO_2}/P_{CH_4})$ is 15.7 (determined with LPGS set-up). Hence, PEG200 is a far less effective additive in increasing the permeability of Pebax® MH 1657 membranes. This test was performed with several blends and included also testing of the $\alpha(P_{CO_2}/P_{He})$ and $\alpha(P_{CO_2}/P_{O_2})$. The data are summarized in Table 4.

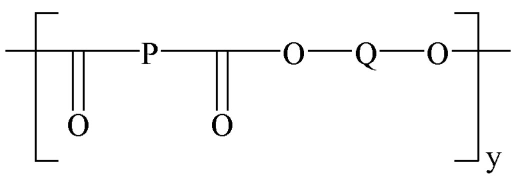
Table 4

Sample	CO_2 permeability	CO_2/light gas selectivity [-]				
	[Barrer]	CO_2/He	CO_2/H_2	CO_2/O_2	CO_2/N_2	CO_2/CH_4
PEBAX®/PEG_{200} blends						
10 wt.% PEG200	127	16.8	10.0	21.3	52.6	16.4
20 wt.% PEG200	136	16.8	10.0	21.0	49.9	15.9
30 wt.% PEG200	153	17.2	10.2	20.9	48.2	15.9
40 wt.% PEG200	159	17.6	10.4	20.8	49.0	15.7
50 wt.% PEG200	172	18.0	10.5	21.0	50.5	15.7

Claims

1. A membrane comprising a polymer composition comprising a blend of a multi-block thermoplastic elastomer comprising alternating hard polymeric blocks and soft polymeric blocks, and a polymer comprising a metal of Group 14 of the Periodic System of the Elements.
2. The membrane according to Claim 1, wherein the hard polymeric blocks are selected from the group consisting of polyurethane blocks, polyamide blocks, non-amorphous polyester blocks, polyimide blocks, polysulfone blocks, polycarbonate blocks and mixtures thereof.
3. The membrane according to Claim 1 or Claim 2, wherein the soft polymeric blocks are selected from the group consisting of polyether blocks, amorphous polyester blocks and polysiloxane blocks.
4. The membrane according to any one of Claims 1 – 3, wherein the multi-block thermoplastic elastomer is a polyamide-polyether multi-block copolymer.
5. The membrane according to Claim 4, wherein the polyamide-polyether multi-block copolymer is selected from the group consisting of:

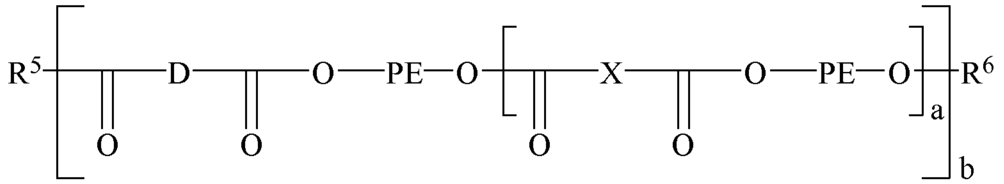
(a) a multi-block thermoplastic elastomer comprising a polyamide block as the hard block and a polyether block as the soft block has preferably the following general Formula (VIII):



Formula (VII)

wherein P is a polyamide block, O is oxygen and Q is a polyether block whereas n indicates an integer such that the intrinsic viscosity η of the multi-block thermoplastic elastomer is in the range of about 0.5 to about 2.5;

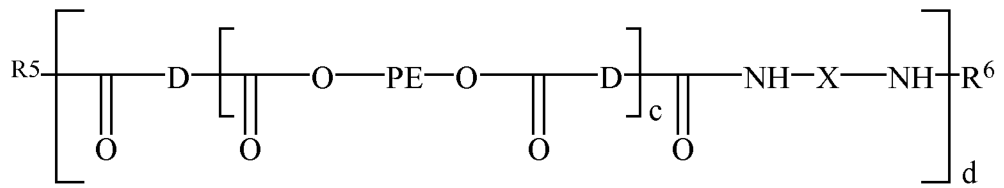
(b) a multi-block thermoplastic elastomer comprising a polyamide block as the hard block and a polyether block as the soft block has preferably the following general Formula (VIII):



Formula (VIII)

wherein R^5 and R^6 are OH and/or H, a is a number 0.1 to 10, b is a number of 2 to 50, D is a residue of an oligoamidediacid or a diacid having a M_n of about 300 to about 8000, PE is a residue of a polyoxyalkylene, preferably a polyoxyethylene, having a M_n of about 200 to about 5000, and X is a residue of a diacid comprising linear or branched aliphatic, cycloaliphatic or aromatic hydrocarbon residues having 4 to 12 carbon atoms; and

(c) multi-block thermoplastic elastomer comprising a polyamide block as the hard block and a polyether block as the soft block has preferably the following general Formula (IX):

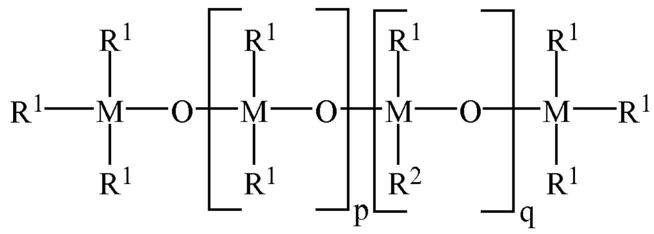


Formula (IX)

wherein R^5 and R^6 are OH and/or H, c is a number 1 to 4, b is a number of 2 to 50, D is a residue of an oligoamidediacid having a M_n of about 300 to about 3000, PE is a residue of a polyoxyalkylene, preferably a polyoxyethylene, having a M_n of about 200 to about 5000, and X is a residue of a diacid comprising linear or branched aliphatic, cycloaliphatic or aromatic hydrocarbon residues having 4 to 20 carbon atoms.

6. The membrane according to any one of Claims 1 – 6, wherein the polymer comprising a metal of Group 14 of the Periodic System of the Elements is selected from the group consisting of:

(a) a polysiloxane according to Formula (I):



Formula (I)

wherein:

M is the Group 14 metal;

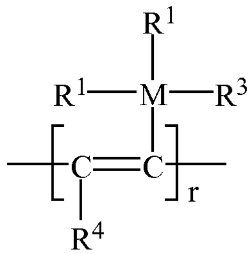
R^1 is selected from the group consisting of linear or branched C_1 - C_{12} alkyl groups, C_6 - C_{12} aryl groups and mixtures thereof;

R^2 is a polyoxyalkylene chain;

p is in the range of 5 to 5000; and

q is in the range of 5 to 10000;

(b) a polyalkyne according to Formula (II):



Formula (II)

wherein:

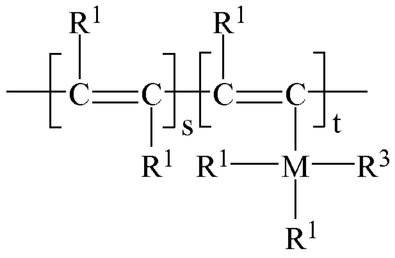
M is the Group 14 metal;

R^1 is selected from the group consisting of linear or branched C_1 - C_{12} alkyl groups, C_6 - C_{12} aryl groups and mixtures thereof;

R^3 is R^1 or a polyoxyalkylene chain;

R^4 is hydrogen, a linear C_1 - C_{12} alkyl group, a $(R^1)_3M\text{-}CH_2\text{-}$ group, a $(R^1)_2R^3M\text{-}$ group, or a mixture thereof; and r is in the range of 100 – 20000;

(c) a polyalkyne according to Formula (III):



Formula (III)

wherein:

M is the Group 14 metal;

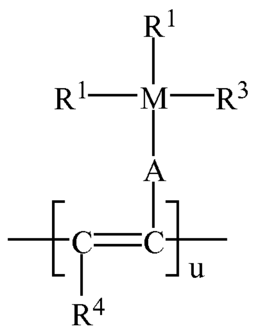
R^1 is selected from the group consisting of linear or branched C_1 - C_{12} alkyl groups, C_6 - C_{12} aryl groups and mixtures thereof;

R^3 is R^1 or a polyoxyalkylene chain;

$(s/(s+t)) \geq 0.2$; and

$(t/(s+t)) \geq 0.05$;

(d) a polyalkyne according to Formula (IV):



Formula (IV)

wherein:

M is the Group 14 metal;

A is $-(CR^1{}_2)_v-$ wherein $v = 1 – 6$;

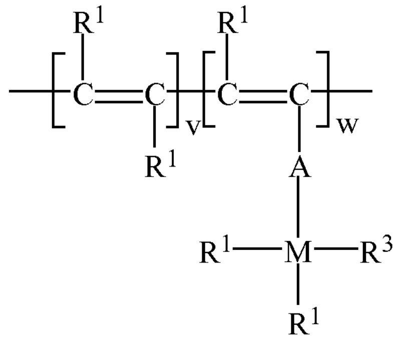
R^1 is selected from the group consisting of linear or branched C_1 - C_{12} alkyl groups, C_6 - C_{12} aryl groups and mixtures thereof;

R^3 is R^1 or a polyoxyalkylene chain;

R^4 is hydrogen, a linear C_1 - C_{12} alkyl group, a $(R^1)_3M\text{-}CH_2$- group, a $(R^1)_2R^3M$- group, or a mixture thereof; and

u is in the range of 100 – 20000;

(e) a polyalkyne according to Formula (V):



Formula (V)

wherein

M is the Group 14 metal;

A is $-(CR^1_2)_v-$ wherein v = 1 – 6;

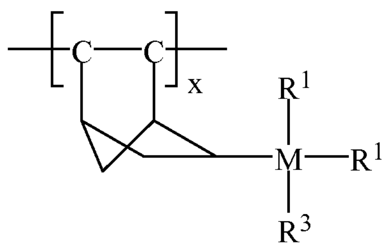
R^1 is selected from the group consisting of linear or branched C_1 - C_{12} alkyl groups, C_6 - C_{12} aryl groups and mixtures thereof;

R^3 is R^1 or a polyoxyalkylene chain;

$(v/(v+w)) \geq 0.2$; and

$(w/(v+w)) \geq 0.05$; and

(f) a polyalkene according to Formula (VI);



Formula (VI)

wherein R^1 is selected from the group consisting of linear or branched C_1 - C_{12} alkyl groups, C_6 - C_{12} aryl groups and mixtures thereof;

R^3 is R^1 or a polyoxyalkylene chain; and

x is in the range of 100 – 20000.

7. The membrane according to Claim 6, wherein the polyoxyalkylene chain is a polyoxyethylene chain.

8. The membrane according to Claim 6 or Claim 7, wherein the weight average molecular weight M_w of the polysiloxane according to Formula (I), the polyalkynes according to Formula (II), (III), (IV) or (V) or the polyalkene according to Formula (VI) is determined by the polyoxyethylene chain for about 60% to about 90%.

9. The membrane according to any one of Claims 6 - 8, wherein the polyoxyethylene chain has the formula $-(CH_2CH_2CH_2O)-[-CH_2CH_2O-]_p-CH_3$, wherein p is in such a range that the M_n of the polyoxyethylene chain is equal to about 1000 or less.

10. The membrane according to any one of Claims 1 - 9, wherein the membrane comprises about 0.1 wt.% to about 99 wt.% of the polymer comprising a metal of Group 14 of the Periodic System of the Elements, based on the total weight of the membrane.

11. The membrane according to any one of Claims 1 - 10, wherein the membrane comprises about 1 wt.% to about 99.9 wt.% of the multi-block thermoplastic elastomer comprising alternating hard polymeric blocks and soft polymeric blocks.

12. The membrane according to any one of Claims 1 - 11, wherein the membrane has a CO_2 gas permeability coefficient of at least of at least 120 Barrer.

13. The membrane according to any one of Claims 1 - 12, wherein the membrane comprises a support.

14. Use of a membrane according to any one of Claims 1 - 13 for separation processes.

15. Use according to Claim 14, wherein the separation process is a gas/gas separation process.

16. Use according to Claim 15, wherein CO_2 is separated from a light gas, wherein the molecular weight of the light gas is 44 or less.

17. Use according to Claim 16, wherein the light gas comprises hydrogen, nitrogen, methane or a mixture thereof.

18. Use of a membrane according to any one of Claims 1 – 13 in ion transport processes.
19. Use of a membrane according to any one of Claims 1 – 13 in ultrafiltration processes and/or nanofiltration processes.

INTERNATIONAL SEARCH REPORT

International application No
PCT/NL2010/050307

A. CLASSIFICATION OF SUBJECT MATTER
INV. B01D71/80 B01D71/70 B01D71/52 B01D71/56
ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)
B01D C08L

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practical, search terms used)

EPO-Internal, WPI Data

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	US 2002/028156 A1 (ANNEAUX BRUCE L [US]; SHALABY SHALABY W [US]; DOOLEY R LARRY [US]) 7 March 2002 (2002-03-07) paragraph [0036] - paragraph [0041]; claims 1-2 the whole document	1-19
X	US 5 552 483 A (HERGENROTHER WILLIAM L [US]; GRAVES DANIEL F [US]) 3 September 1996 (1996-09-03) column 3, line 59 - column 6, line 42; figures column 7, line 55 - column 8, line 7 column 13, line 41 - column 15, line 45; claims 1-7 the whole document	1,3,6-13

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[X] Further documents are listed in the continuation of Box C.

[X] See patent family annex.

* Special categories of cited documents :

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"E" earlier document but published on or after the international filing date

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"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art.

"&" document member of the same patent family

Date of the actual completion of the international search	Date of mailing of the international search report
22 October 2010	03/11/2010

Name and mailing address of the ISA/	Authorized officer
European Patent Office, P.B. 5818 Patentlaan 2 NL - 2280 HV Rijswijk Tel. (+31-70) 340-2040, Fax: (+31-70) 340-3016	Veríssimo, Sónia

C(Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
Y	CAR; ANJA STROPNIK; CHRTOMIR YAVE; PEINEMANN W; K V: "PEG modified poly(amide-b-ethylene oxide) membranes for CO2 separation" JOURNAL OF MEMBRANE SCIENCE, ELSEVIER SCIENTIFIC PUBL.COMPANY. AMSTERDAM, NL, vol. 307, no. 1, 18 December 2007 (2007-12-18), pages 88-95, XP022392367 ISSN: 0376-7388 cited in the application * abstract paragraph [0001] - paragraph [02.1]; tables the whole document	1-19
Y	JIANG X (; DING J; KUMAR A: "Polyurethane-poly(vinylidene fluoride) (PU-PVDF) thin film composite membranes for gas separation" JOURNAL OF MEMBRANE SCIENCE, ELSEVIER SCIENTIFIC PUBL.COMPANY. AMSTERDAM, NL, vol. 323, no. 2, 15 October 2008 (2008-10-15), pages 371-378, XP025677289 ISSN: 0376-7388 [retrieved on 2008-06-28] * abstract paragraph [0001] - paragraph [02.2]; tables paragraph [0005]	1-19
Y	US 5 494 989 A (MILLER WARREN K [US]; FRIESEN DWAYNE T [US]) 27 February 1996 (1996-02-27) cited in the application column 1 - column 2 column 4 the whole document	1-19

INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No
PCT/NL2010/050307

Patent document cited in search report	Publication date	Patent family member(s)	Publication date
US 2002028156 A1	07-03-2002	NONE	
US 5552483 A	03-09-1996	NONE	
US 5494989 A	27-02-1996	DE 19518507 A1 NL 1000434 C2	30-11-1995 24-11-1995